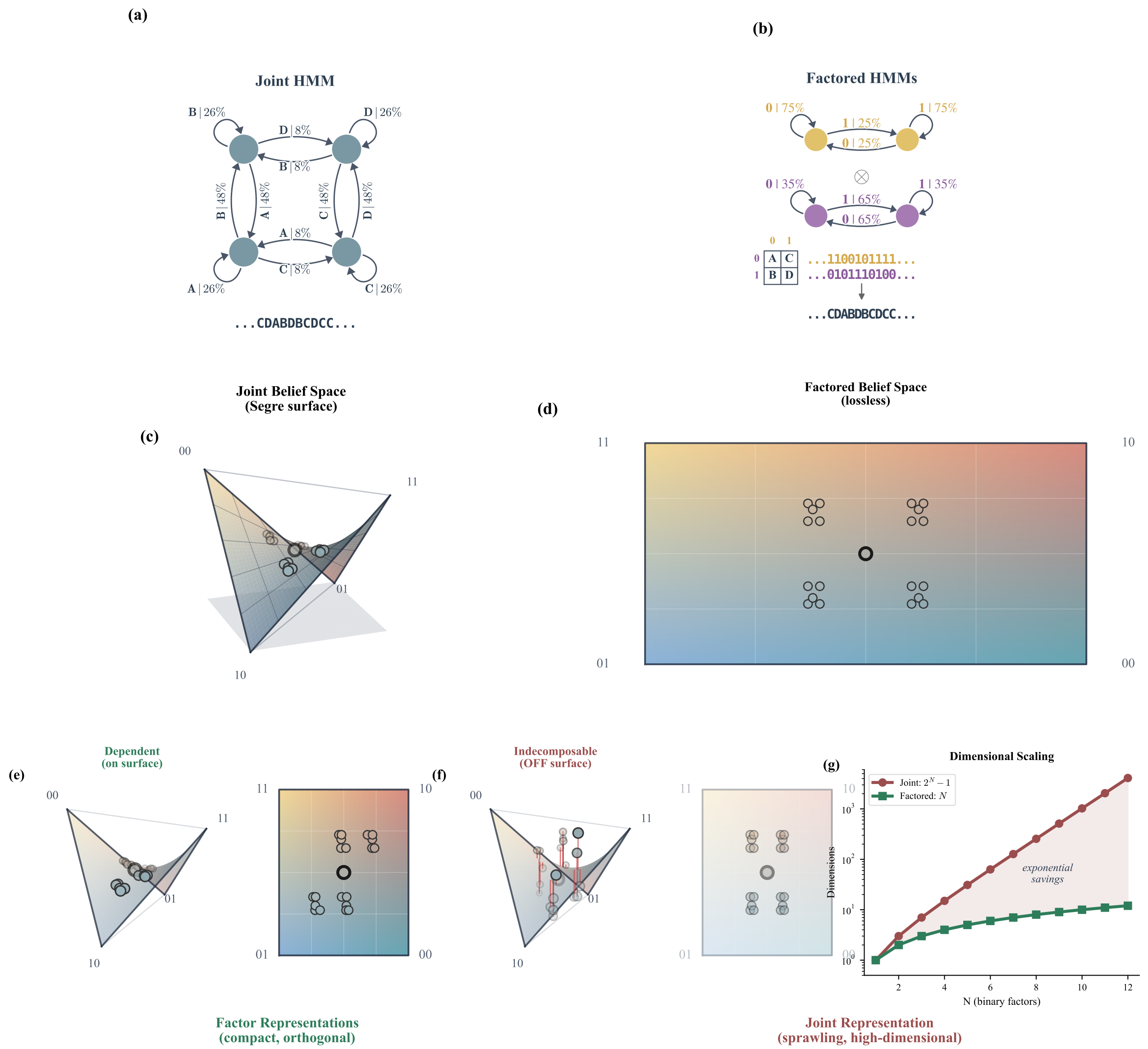
(a)
Joint HMM
B | 26%
D | 26%
D | 8%
B | 8%
B | 48%
A | 48%
C | 48%
D | 48%
A | 8%
C | 8%
A | 26%
C | 26%
...CDABDBCDCC...
(b)
Factored HMMs
0 | 75%
1 | 75%
1 | 25%
0 | 25%
0 | 35%
1 | 35%
1 | 65%
0 | 65%
A C
B D
...1100101111...
...0101110100...
...CDABDBCDCC...
Joint Belief Space
(Segre surface)
Factored Belief Space
(lossless)
(c)
(d)
(e)
Dependent
(on surface)
(f)
Indecomposable
(OFF surface)
(g)
Dimensional Scaling
Joint: $2^N - 1$
Factored: N
exponential
savings
Dimensions
N (binary factors)
Factor Representations
(compact, orthogonal)
Joint Representation
(sprawling, high-dimensional)
(h)